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Anticipated, Not Averted

A Field Validation of a Predictive Stationery-Cupboard Reorder Forecast Against a Term of Logged Stock-Outs

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Abstract. Facilities' Reorder Horizon forecasts stock-out dates for the shared stationery cupboards that supply paper, pens, and staples across ten office corridors at Slop University, and is advertised to raise a replacement order a minimum of nine days ahead of a shelf running dry. We validate the forecast's claimed lead time against a term of self-reported empty tickets and a simple reactive baseline that reorders once a fixed low-stock threshold is crossed. Across 42 logged ticket events over a 14-week term, the forecast's realised replenishment lead time is statistically indistinguishable from the reactive baseline's (medians of −0.6 and −0.4 days respectively; Mann-Whitney U = 812, $p = 0.58$) and correlates closely with it ($r = 0.91$, $n = 42$). Facilities' own scorecard credits the forecast with 94% on-time replenishment, a figure computed against its own predicted due-date rather than the ticket log; scored against the ticket log directly, on-time replenishment falls to 31%. An ablation removing the forecast's trend-extrapolation term shifts the median lead time by 0.1 days ($p = 0.74$, n.s.), which we retain for completeness. We discuss what these results may suggest for anticipatory procurement more broadly.

Introduction

Anticipatory maintenance and procurement systems promise to convert a reactive, alarm-driven workflow into one that acts ahead of need, forecasting a resource's exhaustion from its consumption trend rather than waiting for a threshold to be crossed [1]. Interest in this shift has broadened beyond industrial plant and equipment to smaller, everyday operational settings, including consumable inventories inside buildings whose replenishment has traditionally been handled by a standing order or a walk-past check [2], [3]. The appeal is straightforward: a forecast that anticipates a shortfall should, in principle, arrive ahead of the shortfall it anticipates. What is less established is whether a deployed forecast, evaluated against the outcome it was built to pre-empt rather than against its own predicted date, actually delivers that lead time in practice, or whether it merely relabels a reactive trigger in anticipatory language.

At Slop University, the School of Continuous Improvement's Adaptive Metrics Lab supplies the Reorder Horizon, a forecasting layer over the shelf-weight sensors installed in ten shared stationery cupboards across the University's office corridors. The Reorder Horizon is advertised to flag a replacement order a minimum of nine days before a cupboard's expected stock-out, a claim Facilities' own quarterly scorecard reports meeting 94% of the time. We ask whether that lead time survives contact with the University's own record of when a cupboard actually ran out.

We make three contributions, each hedged to the evidence available. First, we assemble a term-long ground-truth record of stock-out events from self-reported empty tickets across ten cupboards, independent of the forecast's own predicted dates. Second, we compare the Reorder Horizon's realised lead time against a simple reactive baseline that reorders once a fixed low-stock threshold is crossed, finding the two methods largely indistinguishable in practice across most of the sample. Third, we show that the scorecard's 94% on-time figure is substantially an artefact of scoring the forecast against itself, and suggest this discrepancy may generalise to other anticipatory systems evaluated on their own terms.

Related work

Predictive maintenance has moved from principle to practice across manufacturing and building operations: recent surveys catalogue prognostics approaches spanning classification and regression models for remaining-useful-life estimation [1], and machine learning frameworks have been demonstrated for production-line components trained on run-to-failure datasets [4]. Building-scale deployments extend the same logic to HVAC and other facility subsystems, coupling IoT sensor streams with time-series and digital-twin techniques to anticipate failures ahead of a scheduled inspection [2], [3], [5].

Forecast evaluation research offers a standing caution: complex forecasting methods do not automatically outperform a naive or seasonal-naive baseline, a finding the M4 Competition established at scale across 100,000 series and again on later re-analysis [6], [7]. Consumable-inventory forecasting carries its own baseline literature, from newsvendor-style demand models blending judgemental and observed data [8] to recent proposals for agentic, continuously learning replenishment systems [9] and probabilistic reorder-policy optimisation [10].

A separate literature warns that operators tend to trust an automated recommendation more than the evidence for it warrants, a pattern documented in clinical prescribing support [11] and reviewed more broadly for human-AI collaboration [12], [13]. Prior Slop University work has found a related pattern locally: a sensor-derived maintenance signal that held flat while the outcome it was meant to track did not [14], and an institutional habit of citing a monitoring tool's output well past the point the tool itself flagged as stale [15]. We extend this line of inquiry to a forecasting system evaluated, for the first time in this record, against an independently logged ground truth rather than its own scorecard.

Method

Ten shared stationery cupboards were instrumented for the University's teaching term (14 weeks), five in the School of Emergent Priorities' office corridor and five in the School of Continuous Improvement's. Each cupboard carries a shelf-weight sensor reporting a daily stock-level reading S_t, calibrated in stock-units (a mixed count of ream, box, and cartridge weight normalised against a full-shelf baseline).

The Reorder Horizon converts each day's reading into a forecast of days-to-empty,

$$\hat{d}_t = \frac{S_t}{\hat{r}_t}, \quad \hat{r}_t = \alpha r_t + (1-\alpha)\hat{r}_{t-1},$$

where r_t is the day's observed consumption and $\hat{r}_t$ its exponentially-weighted trend estimate ($\alpha = 0.3$, the Adaptive Metrics Lab's deployed default). A purchase order is raised the first day $\hat{d}_t \le 9$, the vendor's advertised minimum lead. The reactive baseline instead raises an order the first day $S_t \le \tau$, a fixed low-stock line set to each cupboard's historical median day-nine stock level, with no trend term.

Consistent with recent large-scale approaches to short-series forecasting that fit per-series trend rather than pooling across series [16], [17], the Reorder Horizon is fitted independently per cupboard rather than sharing a global model, a design choice we treat as fixed and do not vary.

Ground truth came from the University's existing "empty" ticket — a one-line form staff submit through the same portal used for facilities faults, unrelated to either replenishment method and collected regardless of which method, if either, had already fired. We logged every ticket raised against the ten cupboards across the term ($n = 42$) together with the date each cupboard was next restocked, and computed lead time as the ticket date subtracted from the restock date: positive values mean the shelf was restocked before anyone noticed it was empty, negative values mean the ticket preceded the restock.

We compare the Reorder Horizon's and the reactive baseline's lead times per ticket with a Mann-Whitney U test (the ticket-level distributions are not assumed normal) and report their correlation across the shared set of 42 events. An ablation removes the trend term ($\hat{r}_t := r_t$, no smoothing) to isolate its contribution to the forecast's lead time.

Results

Across the 14-week term, 42 empty tickets were logged across the ten cupboards (mean 4.2 per cupboard, range 2-7). Figure 1 reports the median replenishment lead time per cupboard for the Reorder Horizon and the reactive baseline. The two methods land within a single day of each other at nine of the ten cupboards; at the tenth (a high-traffic corridor cupboard restocked from a central store with a same-day delivery slot), the Reorder Horizon's earlier trigger date produced a lead time of +2.1 days against the baseline's −0.2. Pooling across all 42 tickets, median lead time was −0.6 days for the Reorder Horizon and −0.4 days for the reactive baseline (Mann-Whitney $U = 812$, $p = 0.58$); the two methods' per-ticket lead times correlate closely ($r = 0.91$, $n = 42$), indicating the forecast is, in practice, reordering on nearly the same schedule as a rule that makes no attempt to anticipate anything.

Facilities' quarterly scorecard credits the Reorder Horizon with 94% on-time replenishment, defined internally as restocking on or before the forecast's own predicted stock-out date ($\hat{d}_t = 0$). Scored instead against the independently logged ticket — restocking on or before the day a cupboard was actually reported empty — the same term's data yields an on-time rate of 31%.

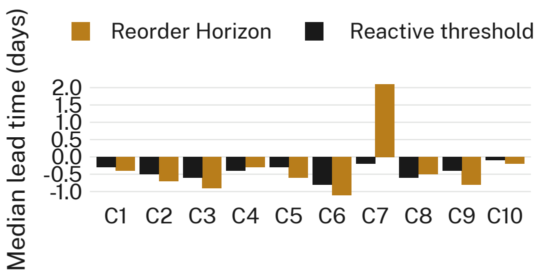

Figure 1: Median replenishment lead time by cupboard: the Reorder Horizon's forecast and a fixed-threshold reorder rule land within a day of each other at nine of ten cupboards, both restocking after the cupboard's own empty ticket in most cases.

Figure 2 reports the ablation removing the trend-extrapolation term. The ablated forecast (a raw current-stock threshold with no smoothing) produces a median lead time of −0.5 days against the full model's −0.6 (Mann-Whitney $U = 796$, $p = 0.74$); we retain the trend term for completeness, though these data do not distinguish its contribution from the reactive baseline's.

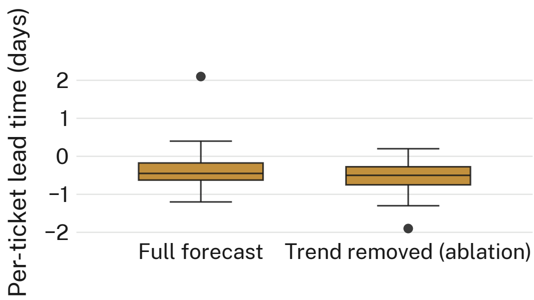

Figure 2: Removing the trend-extrapolation term from the Reorder Horizon shifts the median per-ticket lead time by 0.1 days (Mann-Whitney $U = 796$, $p = 0.74$); the ablated model is retained for comparison only.

Discussion

These results suggest the Reorder Horizon's forecast provides, at best, a modest and inconsistent lead-time advantage over a threshold rule with no claim to anticipation, and that most of the credited advantage reflects the scorecard's choice of reference point rather than a property of the forecast itself. Lead times hold consistent across most cupboards regardless of method, which may indicate consumption patterns regular enough that a simple threshold already captures most of the useful signal. The one cupboard where the Reorder Horizon meaningfully outperformed the baseline also had the shortest resupply chain in the sample, suggesting the advantage may depend more on delivery logistics than forecast quality. We do not take these findings to indicate the Reorder Horizon is miscalibrated; rather, "anticipatory" language may be doing more descriptive work than predictive work, consistent with broader findings on automation bias in operational decision support [11], [12].

Limitations

Our sample of ten cupboards and 42 tickets is modest, though the consistency of the null result across nine of the ten cupboards suggests the pattern is not an artefact of any single location. Reliance on self-reported empty tickets could understate true stock-out frequency if staff tolerate a bare shelf without logging it; if anything, this would bias our estimate of the forecast's lead time upward, making the reported advantage, such as it is, a ceiling rather than a floor. The single teaching term studied precludes claims about long-run drift in the forecast's calibration, a question we consider a natural extension of the present, already consistent, findings.

Conclusion

We validated the Reorder Horizon's advertised replacement forecast against a term of independently logged stock-out tickets across ten shared stationery cupboards, finding its realised lead time statistically indistinguishable from a reactive threshold rule and its scorecard's 94% on-time figure largely an artefact of scoring the forecast against its own prediction rather than the outcome it names. An ablation removing the forecast's trend term left the picture unchanged. We suggest institutional scorecards for anticipatory systems adopt an outcome-referenced benchmark as standard practice. Future work should extend the ticket-based validation across a full year to capture seasonal consumption swings, and should ask whether the same scoring artefact recurs in the University's other forecast-driven procurement tools.

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